AN INTELLIGENT HOSTEL OPERATIONS, SAFETY AND RESOURCE OPTIMIZATION PLATFORM

ABSTRACT

An intelligent and unified hostel management platform is proposed to improve the efficiency, safety, and resource utilization of hostel operations. Conventional hostel management systems primarily focus on administrative activities and often operate as fragmented modules with limited intelligence and decision-support capabilities. The proposed system addresses this limitation by integrating hostel administration with artificial intelligence, predictive analytics, real-time monitoring, and resource optimization. The proposed system provides centralized management of students, rooms and beds, leave requests, entry and exit movements, fees, complaints, mess operations, inventory, maintenance, visitors, and emergency incidents. Role-based access control enables secure interaction for students, parents or guardians, wardens, security personnel, administrators, and mess or maintenance staff. The platform incorporates AI-based complaint classification, explainable room allocation, predictive maintenance, meal-demand forecasting, inventory forecasting, and risk analysis. An emergency management module supports SOS reporting, alerts, acknowledgement, tracking, and audit logging. The primary objective is to transform hostel management from basic record keeping into intelligent monitoring, prediction, optimization, and decision support. The proposed system is expected to improve operational efficiency, resource utilization, safety, complaint handling, meal planning, maintenance management, and administrative decision-making.

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